

Creative Core -- Worksheet 2

Animation + VFX Practice Scene

Task: Build a scene that demonstrates core animation and VFX skills. You will create in-editor animations, set up an Animator Controller with transitions, and add particle effects. The scene does not need to be a game -- it just needs to showcase each concept clearly. Check off each requirement as you complete it.

Assets: Use any combination of: *Creative Core starter files, Junior Programmer prototype assets (vehicles, animals, food), Unity primitives, or free Asset Store models. A cube that bounces and a sphere that pulses count -- focus on the animation and VFX concepts, not art.*

Setup

- [] **New 3D project** (Built-in RP or URP). Name it *AnimVFX_Practice*. Create folders: Assets/Animations, Assets/Scripts, Assets/Prefabs.
- [] **Scene layout:** Place at least 3 objects in a simple scene (ground plane + objects). Import any assets you want to animate.

Part 1 -- Animation

A. In-Editor Keyframe Animations

Create animations directly in Unity's Animation Editor using Record Mode and keyframes.

- [] **Animation Clip 1 -- Looping motion:** Select an object, open Animation Editor (Window > Animation > Animation), create a clip that loops continuously. Animate at least **2 properties** (e.g., position + scale, or rotation + color). Minimum 3 keyframes.
- [] **Animation Clip 2 -- One-shot action:** On a *different* object, create a clip that plays once (uncheck Loop Time on the .anim asset). Example: a door opening, a platform rising, an object scaling up then staying.
- [] **Curves refinement:** On at least one clip, switch to the **Curves Editor** (bottom of Animation Editor) and adjust the interpolation curves. Make the motion feel intentional -- ease in, ease out, or sharp impact.

B. Animator Controller + State Machine

Set up an Animator Controller that transitions between states based on a parameter.

- [] **Animator Controller** with at least **3 states** visible in the Animator window (Window > Animation > Animator). One must be the default state (orange).
- [] **At least 1 Bool or Trigger parameter** in the Parameters tab that drives a transition.
- [] **Transitions configured:** At least 2 transitions with conditions. At least one transition must have **Has Exit Time** unchecked (immediate trigger).
- [] **Script-driven transition:** Write a script that calls **animator.SetBool()** or **animator.SetTrigger()** based on player input (key press) or a trigger collider (OnTriggerEnter). Attach it to the appropriate object.

Hint: Remember that you can always import animations from asset stores and create ones easily using some websites for your projects.

Part 2 -- VFX (Particle Systems)

C. Environmental Particle System (Continuous)

Create a continuous particle effect from scratch (no prefabs -- build it yourself).

- [] **New Particle System** (Hierarchy > Effects > Particle System). Rename it descriptively (e.g., FX_Snow, FX_Dust, FX_Embers).
- [] **Main module configured:** Looping = on, Prewarm = on, Start Lifetime / Start Speed / Start Size / Max Particles all set to intentional values (not defaults).
- [] **Shape module:** Change from default Cone to a shape that fits your effect (Box for weather, Sphere for ambient dust, etc.). Scale the shape to cover the desired area.
- [] **Emission module:** Rate over Time set to a value that looks good without tanking performance.
- [] **Renderer module:** Assign a material/texture that fits the effect (snowflake, raindrop, ember, dust). Change Render Mode if needed (Stretched Billboard for rain).
- [] **At least 1 additional module enabled:** Color over Lifetime, Size over Lifetime, or Noise. The effect must visually evolve over each particle's lifetime -- not just static dots.

D. Burst Particle System (One-Shot)

Create a one-shot burst effect triggered by player input or a collision.

- [] **New Particle System.** Looping = off, Play On Awake = off. Rename it (e.g., FX_SmokePuff, FX_Explosion, FX_Confetti).
- [] **Emission:** Rate over Time = 0. Add a **Burst** (click + under Bursts) with Count = 15-50 at Time = 0.
- [] **Color over Lifetime:** Particles fade to transparent by end of life (use gradient editor).
- [] **Size over Lifetime:** Particles grow or shrink over their lifetime (use curve editor).
- [] **Triggered from script:** Write a script (or reuse the tutorial's ToggleFireParticle) that calls **particleSystem.Play()** on a key press or OnTriggerEnter event. The effect must NOT play on scene start.

Submission

Submit: Zipped Unity project folder **or** GitHub repository link.

References: Creative Core Animation mission (learn.unity.com) | Creative Core VFX mission (learn.unity.com)